

[Turn over

1 Real numbers are stored in a computer system using floating-point representation with:

- 10 bits for the mantissa
- 6 bits for the exponent
- two's complement form for both the mantissa and the exponent.

(a) Calculate the denary value of the given normalised floating-point number.

Show your working.

Mantissa

0	1	0	0	1	1	1	1	0	0
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Exponent

0	0	1	0	0	1
---	---	---	---	---	---

Working

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Answer

[3]

(b) Calculate the normalised floating-point representation of -102.75 in this system.

Show your working.

Mantissa

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Exponent

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Working

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- 2 The TCP/IP protocol suite has four layers:

Transport, Application, Link, Internet

- (a) Complete the diagram to show the correct order for these layers.

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- (b) Describe the function of the Transport layer.

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- (c) Outline **one** protocol that is associated with the Application layer.

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- 3 (a) Explain what is meant by **non-composite** and **composite** data types.

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(b) Write **pseudocode** statements to declare the record data type `FootballClub` to hold data about football clubs in a league, to include:

- name of team
- date team joined the league
- main telephone number
- name of the manager
- number of members
- current position in the league.

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4 (a) Describe the sequential method of file **access**.

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(b) Explain how the sequential method of file access is applied to files with serial organisation and to files with sequential organisation.

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- 5 (a) Write this Reverse Polish Notation (RPN) in infix form:

5 2 + 9 3 - / 3 *

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- (b) Write this infix expression in RPN:

$((7 + 3) - (2 * 8)) / 6$

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- (c) Evaluate this RPN expression:

a b - c d + * e /

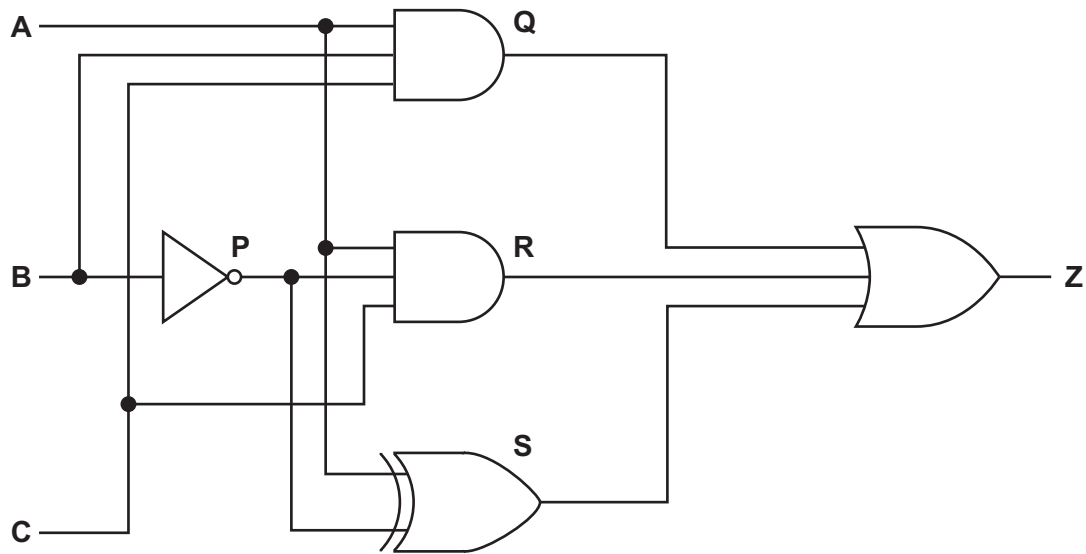
when

a = 17, b = 5, c = 7, d = 3 and e = 10

Show the changing contents of the stack as the RPN expression is evaluated.

[4]

6 The diagram shows a logic circuit.



(a) Complete the truth table for the given logic circuit.

Show your working.

			Working space				
A	B	C	P	Q	R	S	Z
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

[3]

(b) Write the Boolean expression that corresponds to the logic circuit as a sum-of-products.

Z =

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- (c) (i) Complete the Karnaugh map (K-map) for the Boolean expression:

$$\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + A.B.\bar{C} + A.B.C$$

		BC			
		00	01	11	10
A	0				
	1				

[2]

- (ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]
- (iii) Write the Boolean expression from your answer to part (c)(ii) as a simplified sum-of-products.

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- 7 (a) Describe what is meant by a digital certificate.

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- (b) Explain the role of a digital certificate in creating a digital signature.

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- 8 A **declarative** programming language is used to represent the features that are available and the features that are unavailable on different body styles of a car.

```

01 feature(sunroof).
02 feature(automatic_tailgate).
03 feature(heated_seats).
04 feature(extra_seats).
05 feature(reversing_camera).
06 feature(dashboard_camera).
07 feature(air_conditioning).
08 feature(heated_windscreen).
09 feature(satnav).
10 bodystyle(saloon).
11 bodystyle(hatchback).
12 bodystyle(estate).
13 bodystyle(minivan).
14 bodystyle(convertible).
15 available(sunroof, hatchback).
16 available(sunroof, minivan).
17 available(reversing_camera, hatchback).
18 available(extra_seats, minivan).
19 available(reversing_camera, saloon).
20 unavailable(sunroof, convertible).
21 unavailable(automatic_tailgate, saloon).
22 unavailable(extra_seats, hatchback).

```

These clauses have the meanings:

Clause	Meaning
01	Sunroof is a feature.
10	Saloon is a body style.
15	Sunroof is available on a hatchback.
20	Sunroof is unavailable on a convertible.

- (a) Sliding doors is a feature that is available on a minivan but unavailable on a hatchback.

Write additional clauses to represent this information.

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[3]

(b) Using the variable `Options`, the goal:

```
available(Options, saloon)
```

returns

```
Options = reversing_camera
```

Write the result returned by the goal:

```
available(Options, hatchback)
```

`Options = .....` [1]

(c) `F` may be available for `B` if `F` is a feature and `B` is a body style and `F` is **not** unavailable for that body style.

Write this as a rule:

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may_choose_option(F, B)
```

IF

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9 Explain what is meant by **Deep Learning** in relation to Artificial Intelligence (AI).

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- 10 (a) State a condition that must be true for an array to be searchable for a binary search.

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- (b) Complete the given pseudocode to find an item in a 1D array `Names` of type `STRING` using a binary search.

```

DECLARE Names : ARRAY[1:100000] OF STRING
DECLARE TopOfList : INTEGER
DECLARE EndOfList : INTEGER
DECLARE CurrentItem : INTEGER
DECLARE ToFind : STRING
DECLARE Found : BOOLEAN
DECLARE NotInList : BOOLEAN
TopOfList ← 1
EndOfList ← 100000

OUTPUT "Which name do you wish to find? "
INPUT ToFind

.....

NotInList ← FALSE

WHILE ..... AND .....
    CurrentItem ← (TopOfList + EndOfList) DIV 2

    IF ..... THEN
        Found ← TRUE
    ELSE
        IF TopOfList >= EndOfList THEN
            .....
        ELSE
            IF ToFind > Names[CurrentItem] THEN
                .....
            ELSE
                EndOfList ← CurrentItem - 1
            ENDIF
        ENDIF
    ENDIF
ENDWHILE
IF Found = TRUE THEN
    OUTPUT "Item found at position ", CurrentItem, " in array"
ELSE
    OUTPUT "Item not in array"
ENDIF
  
```

[5]

- (c) Describe the performance of a binary search in relation to the number of data items in the array being searched. Refer to Big O notation in your answer.

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- 11 Reduced Instruction Set Computers (RISC) and Complex Instruction Set Computers (CISC) are two types of processor.

- (a) State **two** features of RISC processors.

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- (b) Outline the process of interrupt handling as it could be applied to RISC or CISC processors.

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- (c) Explain how pipelining affects interrupt handling for RISC processors.

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